MMM-A-002015(GSA-FSS)
April 6, 1977

INTERIM FEDERAL SPECIFICATION

ADHESIVE, SEMI-SOLID, STICK FORM, WITH DISPENSERS

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington DC 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 This specification covers a convenience type, quick setting office adhesive in stick form, suitable for bonding paper, photographs, and fabric.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

UU-P-121 - Paper, Bond and Writing; White and Colors.
C-C-116 - Card, Tabulating.

Federal Standards:

Federal Standard 123 - Marking for Shipment (Civil Agencies).
Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer, and Related Materials, Methods of Inspection, Sampling, and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8040

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Laws and Regulations:

16 CFR 1500 - Hazardous Substances and Articles, Administration and Enforcement Regulations.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 903 - Peel or Stripping Strength of Adhesive Bond.
D 1833 - Odor of Petroleum Wax.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Technical Association for Pulp and Paper Industry (TAPPI):

Suggested Method T 517 - Dynamic Strength of Flexible Barrier Material Seals.

(Application for copies should be addressed to Technical Secretary, TAPPI, 360 Lexington Avenue, New York, NY 10017.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Unit containers. The semi-solid, white or clear adhesive stick shall be housed in a rigid vapor-tight closable container in such a way that the stick can be mechanically advanced to a protruding position, and mechanically retracted without making contact with the adhesive stick, as required for application and storage. The unit container shall have a cap which shall form a vapor-tight seal when pressed to a closed position on the container. The container and cap shall be made of impact-resistant material which will neither deteriorate nor cause deterioration of the adhesive (see 5.1).

3.2 Nonvolatile content. When tested as specified in 4.4.2, the product shall have a minimum nonvolatile content of 40 percent.

3.3 Set time. When tested as specified in 4.4.3, the product shall produce the minimum fiber delamination in two out of three samples. The minimum fiber delamination shall be 90 percent of each of 8 out of 10 specimens.

3.4 Wrinkle, shrink, and curl. When tested as specified in 4.4.4, the product shall not cause the test paper to wrinkle, shrink, or curl.

3.5 Mold growth. None of the adhesive-covered test specimens shall support mold growth in excess of that of the control specimen when tested as in 4.4.1.

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3.6 Discoloration of paper and photographs. When tested as specified in 4.4.5, the product shall cause no discoloration of paper or photographs.

3.7 Stability of the container. When tested as specified in 4.4.6, the container shall meet all requirements specified in 3.1.

3.8 Loss of weight. When tested as specified in 4.4.7, the product shall not lose weight in excess of one percent.

3.9 Odor. When tested as specified in 4.4.12, the odor of the adhesive shall not be greater than 1.0.

3.10 Workability. When applied as specified in 4.4.4, the adhesive shall be uniform, even-spreading, and free of dirt, lumps, foreign matter, coarse particles, and all other defects which would adversely affect its useability. The adhesive stick shall be hard enough to prevent an excessive use rate, messy and inaccurate application, difficulties in retracting the stick, and transfer of the adhesive to the outside of the cap or of the container when the cap is replaced. The stick shall not be so hard as to prevent easy application.

3.11 Stability of the adhesive. After being stored as specified in 4.4.6, the product shall meet the requirements of 3.3 and 3.10, and show no change in its infrared spectrum when the spectrum is obtained and examined as specified in 4.4.8.

3.12 Adhesion strength. When tested as specified in 4.4.9.1, the paper shall break rather than the bond in 9 out of 10 specimens. The adhesive shall have a minimum peel strength of 0.5 lbs/inch, when tested as specified in 4.4.9.2.

3.13 Shelf storage life. After storage as specified in 4.4.10, the adhesive shall conform to the requirements of 3.11.

3.14 Toxicity. When tested as specified in 4.4.11, the adhesive shall produce no acute dermal toxicity, no positive eye irritant reaction, and shall have a primary irritation (dermal) score of 0.5 or less.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection and testing of the end item.

4.2.1 Lot. The adhesive shall be assembled into lots as specified in MIL-STD-105. In MIL-STD-105, the words "essentially the same conditions" shall be interpreted to mean all filled unit containers of the same size offered for delivery at one time.

4.2.2 Sampling. For the purposes of sampling, the lot shall be expressed in unit containers. Samples from lots shall be taken in accordance with MIL-STD-105 using inspection level S-2 and an AQL of 2.5 percent defective.

4.2.3 Inspection. Unit containers shall be operated to ensure that the adhesive stick can be advanced and retracted as specified in 3.1. After being dropped 5 times in random orientations from a height of 1 m to a concrete floor, each unit container shall show no chips, cracks, or dents visible to the unaided eye, and shall meet all requirements specified in 3.1.

4.2.4 Certificate of compliance. The supplier shall submit to the contracting officer a certificate of compliance indicating that the adhesive formula meets the toxicity requirements specified in 3.14, and that the adhesive complies with the shelf storage life requirements specified in 3.13. When a certificate of compliance is submitted, the government reserves the right to test such items to determine the validity of the certificate.

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4.3 Inspection of Preparation for Delivery.

4.3.1 Inspection of containers. Each container randomly selected from the lot offered for inspection shall be examined for defects of construction of the container and the closure, for evidence of leakage, and for unsatisfactory markings; each container shall also be tested to determine the amount of contents. Any container in the sample having one or more defects or under required fill shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, the lot represented by the sample shall be rejected.

4.3.2 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements of section 5. Defects shall be scored in accordance with table I. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, and selected at random just prior to the closing operations. Defects listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-4 and the AQL shall be 4.0 defects per hundred units.

TABLE I. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.
Contents (exterior container)	Number per container is more or less than required. Net weight exceeds requirements.

4.4 Test methods. All tests shall be conducted in accordance with the methods specified. Unless otherwise specified, all tests shall be performed at standard testing conditions which are 23 ± 1°C (73 ± 2°F) and a relative humidity of 50 ± 5 percent. All test reports shall contain the individual values utilized in obtaining the final result. Each final result shall be compared with the applicable requirement in section 3 to determine compliance with this specification. Failure to pass any test shall be cause for rejection of the sample and therefore, the lot.

4.4.1 Mold growth. The adhesive shall be tested for conformance to 3.5 as specified in Method 6271 of Fed. Test Method Std. No. 141, using the following procedures:

(a) Three test specimens shall be prepared by applying an even coat of adhesive to both sides of 32 mm or larger squares of glass. These specimens shall be innoculated with mold species as specified.

(b) One control specimen shall consist of a clean glass square innoculated in the same way.

(c) After seven days of storage as specified, the test specimens shall be compared with the control specimen for the amount of mold growth.

(d) Viability controls shall be used as specified to determine the viability of the innoculum.

4.4.2 Nonvolatile content. A vapor-tight container and matching cap shall be dried overnight in a desiccator and then weighed. A 20-g portion of the sample shall be added to the container, and the container shall be closed and weighed. Next, the container shall be opened, placed with its cap in an oven at 105°C for six hours, cooled in desiccator, and then weighed with the cap. The heating and weighing procedure shall be continued by heating for periods of 1/2 hours until the weight remains constant. The nonvolatile content shall be calculated for conformance to 3.2.

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4.4.3 Set time. Ten specimens shall be prepared by cutting paper conforming to G-C-116 into 25-by 150-mm strips and applying 60 ± 10 mg of adhesive to a 25-by 125-mm end. In applying the adhesive, sufficient strokes shall be used to obtain an even coat. Immediately after applying the adhesive to one strip, a second clean strip shall be bonded to it so that the non-bonded portions face each other at the same end. Bonding shall be achieved by using finger pressure to rub the top strip down on the adhesive covered strip for ten seconds. After a 15-minute set time, each specimen shall be laid on a table, and the top strip shall be pulled back at a 180° angle. Strips shall be examined for fiber delamination for conformance with 3.3.

4.4.4 Wrinkle, shrink, curl, and workability. An even film of adhesive shall be applied to a 60- by 90-mm area approximately in the center of a 75-by 100-mm sheet of paper conforming to UU-P-121, type III. During the application, the adhesive shall be examined for the workability properties specified in 3.10. During the application, and during and after drying, the specimens shall be examined for wrinkle, shrink, or curl for conformance with 3.4.

4.4.5 Discoloration of paper and photographs. Adhesive shall be applied to white paper conforming to UU-P-121, type III, and to the back of a type 107 Polaroid print. After seven days at standard conditions, the paper and photographs shall be examined for discoloration as specified in 3.6.

4.4.6 Stability of the container. The adhesive, in the original unit container, shall be stored in an oven at 49°C for 14 days. The adhesive shall be examined as specified in 4.4.8. The container shall be examined as specified in 3.7.

4.4.7 Loss of weight. The adhesive in the original unit container, shall be weighed, and then stored for 14 days at 49°C. Next, the specimen shall be weighed, the adhesive shall be removed from the container, and then the container alone shall be weighed. The percentage loss of weight of the adhesive shall be calculated for conformance with 3.8.

4.4.8 Stability of the adhesive. The adhesive shall be applied to a teflon sheet, allowed to dry, and then peeled off. This specimen shall be placed in an infrared spectrophotometer, and its spectrum shall be recorded over the range of 4,000 to 625 cm^{-1}. After storage as specified in 4.4.6, the spectrum shall be obtained in the same manner and examined for differences as specified in 3.11.

4.4.9 Adhesion strength.

4.4.9.1 Tensile "lap" seal strength. The product shall be tested for conformance with 3.12 as specified in TAPPI suggested method T 517 using the following procedure:

(a) The adhesive shall be applied to strips made from paper conforming to G-C-116 with sufficient strokes to form an even coat. A clean strip shall be bonded to a strip prepared with adhesive as shown in T 517, figure 2, using tight finger pressure for ten seconds.

(b) Ten specimens shall be conditioned for 24 hours at standard conditions, and tested at a crosshead speed of 125 mm per minute. Specimens shall be examined to determine whether the failure was in the paper or the bond.

4.4.9.2 Peel strength. The product shall be tested for conformance with 3.12 as specified in ASTM D 903, using the following procedure:

(a) The material used shall be 100 percent cotton jeans fabric, bleached, lint-free, 4-1/2 oz/yd^2, 95 x 64 gray fabric.

(b) Specimens shall be conditioned for three days at standard conditions.

(c) Specimens shall be prepared using sufficient strokes of the adhesive stick to apply an even coat weighing 70 ± 10 mg to each of the strips to be bonded. The bond shall be made using finger pressure for 15 seconds to rub the top strip down on the bottom strip.

(d) Ten samples shall be tested at a crosshead speed of 300 mm per minute.

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4.4.10 Shelf storage life. One unit container of the product shall be stored for one year from the date of manufacture at standard conditions. The adhesive shall be tested for the requirements of 3.11.

4.4.11 Toxicity. The adhesive shall be tested for conformance to 3.14 using the following procedures as specified in 16 CFR 1500, Hazardous Substances and Articles, Administration and Enforcement Regulations:

(a) Method of testing toxic substances (acute dermal toxicity).
(b) Method of testing primary irritant substances.
(c) Test for eye irritants.

4.4.12 Odor. The adhesive shall be tested for conformance to 3.9 as specified in ASTM D 1833. Samples shall be prepared by transfering approximately 10g of the adhesive in thin shavings to clean odor-free bottles.

5. PREPARATION FOR DELIVERY

5.1 Unit containers. Unless otherwise specified, the net weight of adhesive in each unit container shall be 0.22 to 0.27 oz., 0.40 to 0.60 oz., or 1.1 to 1.5 oz.

5.2 Packaging. Packaging shall be level A or commercial as specified (see 6.2).

5.2.1 Level A. The adhesive shall be packaged in unit containers conforming to PPP-B-566, variety 2, style optional.

5.2.2 Commercial. The adhesive shall be packaged in unit containers normally used in commercial practice. The complete package shall be designed to protect the item against damage during shipment, handling, and storage.

5.3 Packing. Packing shall be level A or commercial as specified (see 6.2).

5.3.1 Level A. The adhesive in unit containers shall be packed in boxes conforming to PPP-B-636, V3c. Closure and sealing shall be in accordance with the appendix to the box specification.

5.3.2 Commercial. The adhesive shall be packed in containers that insure safe delivery at destination, provide for safe redistribution by the initial receiving activity, and are acceptable by common carrier under National Motor Freight Classification or Uniform Freight Classification.

5.4 Unitization. (Applicable to full rail car and truck load shipments only.) The adhesive shall be unitized for shipment in accordance with normal commercial practice. The unitized load shall not exceed 2,500 lbs in weight, 65 inches in height, 56 inches in length, and 45 inches in width.

5.5 Marking. Packages, shipping containers and unitized loads (when applicable) shall be marked in accordance with Fed. Std. No. 123.

5.6 Special marking. In addition to other marking required by 5.5, each shipping container and intermediate package, as applicable, shall be marked as follows:

(a) Date of manufacturer (by month and year, not by code).
(b) Date of first reinspection (insert date, by month and year, one year after date of manufacture).

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6. NOTES

6.1 Intended use. This convenience-type semi-solid adhesive is intended for mounting photographic prints, maps, drawings, and charts. The adhesive is also suitable for bonding fabrics in cases where high strength is not required.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Quantity required and size of container (see 5.1).
(c) Level of packaging and packing required (see 5.2 and 5.3).
(d) Unitization if applicable (see 5.4).
(e) Special marking required (see 5.6).

GSA DC 77-4661

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